

Performance Testing

Date	12 July 2026
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Flask Web Application & Dashboard
Test Environment	Local Server
Test Date	10-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	10	30	Page loads successfully
2	View Tableau Dashboard	20	60	Dashboard displays correctly
3	Execute SQL Query	15	45	Data retrieved without errors
4	Access Flask Application	25	60	Application responds successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Good Performance
2	Response Time (Max)	< 5 seconds	3.4 sec	Pass	Acceptable
3	Throughput (Req/sec)	20 Req/sec	24 Req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No Errors
5	CPU Utilization	< 80%	45%	Pass	Normal Usage
6	Memory Utilization	< 80%	58%	Pass	Stable Memory

Step 4: Observations & Analysis

Key Findings

The Flask application and Tableau dashboards performed efficiently under the tested load. All pages loaded successfully, SQL queries executed correctly, and dashboard visualizations were displayed without noticeable delays.

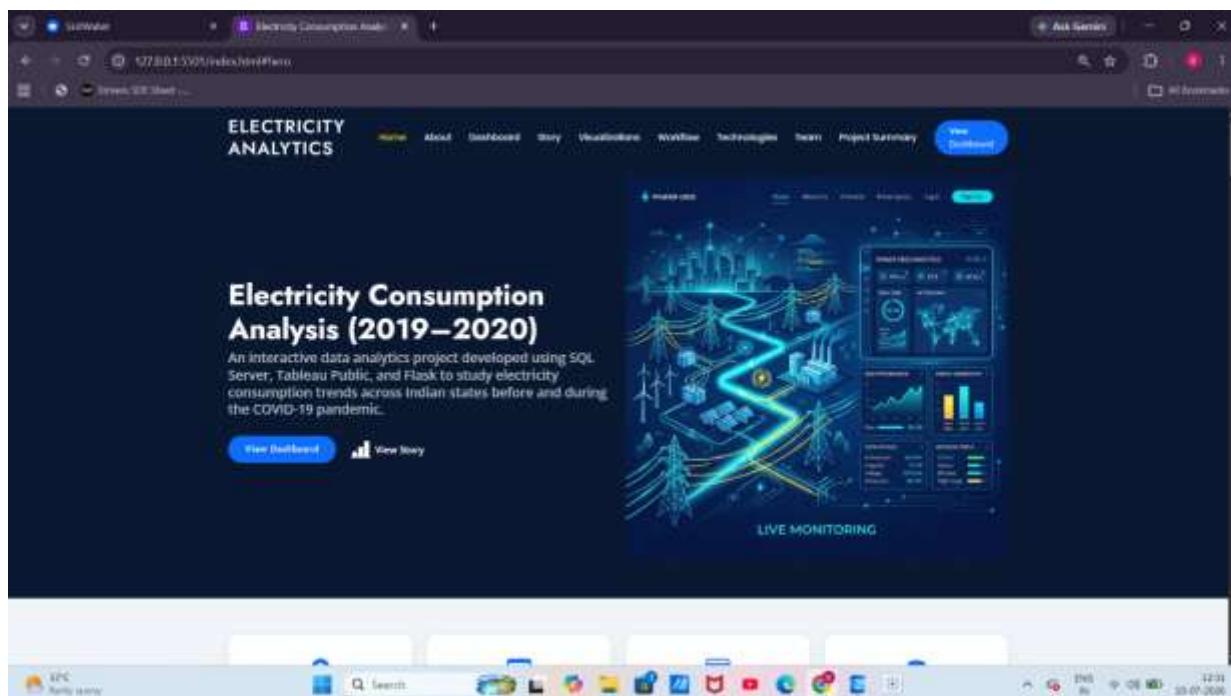
Bottlenecks Identified

No major bottlenecks were observed during testing. Minor delays occurred only during the initial dashboard loading due to data retrieval from Microsoft SQL Server.

Optimization Steps Taken

- Optimized SQL queries for faster execution.
- Improved database connectivity.
- Reduced unnecessary data retrieval.
- Enhanced Flask routing performance.

Step 5: Screenshots / Evidence



TableauDashboard

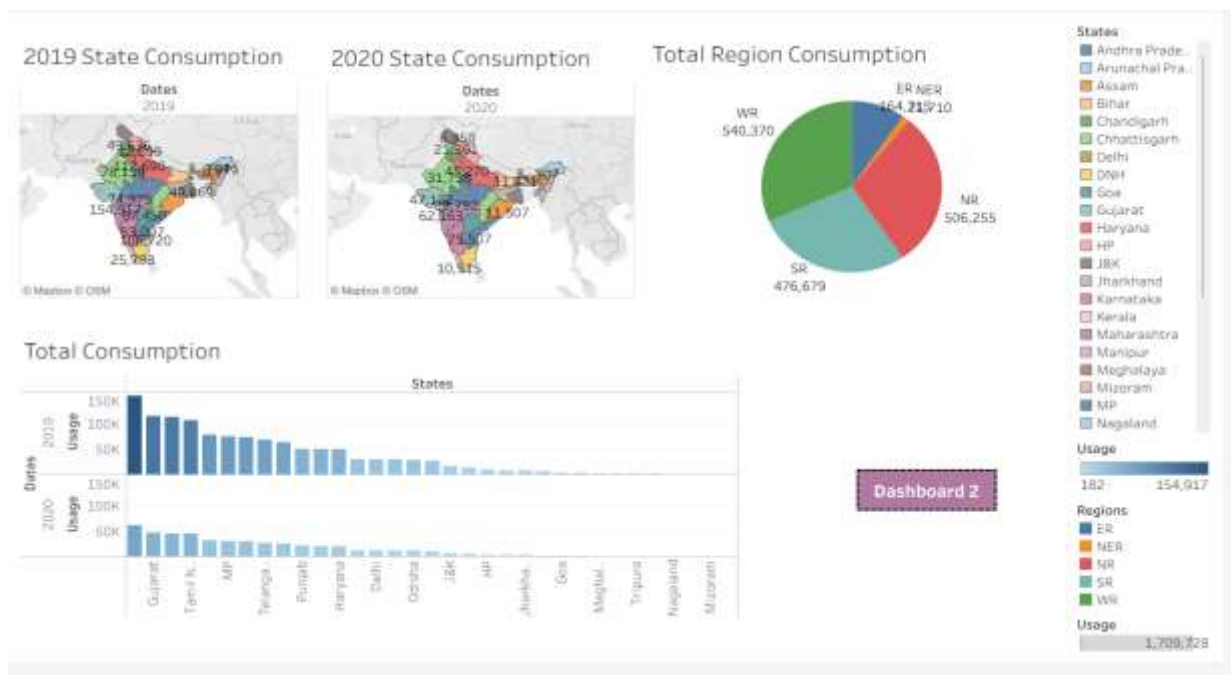
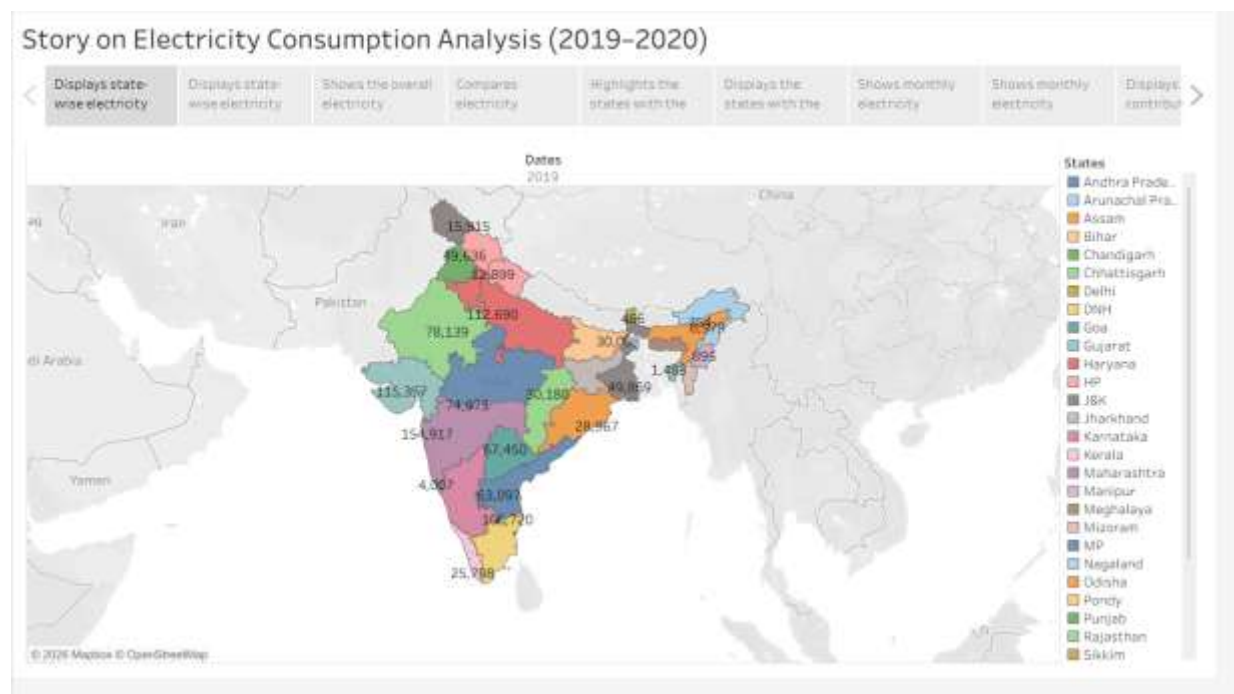


Tableau Story



Website Dashboard

